Alan Karp

Director, Business Operations and AI Enablement

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Director of Business Operations and AI Enablement with 10 years driving enterprise transformation at PE-backed manufacturing platforms. At DwyerOmega, led cross-functional program execution across ERP/MRP, AMS, CRM/MarTech, and enterprise BI through 6 acquisitions ($800M+), reporting to PE stakeholders and executive leadership throughout. Currently at Syrup Labs AI, building and deploying agentic AI systems, lead capture workflows, and automation infrastructure for small businesses, translating messy operational realities into structured, repeatable automation programs. Manufacturing specialist: ERP/MRP integration, M&A consolidation, and AI-assisted knowledge management across industrial operations. MBA, Villanova (Strategy, Analytics, AI/ML). B.S. Chemical Engineering, UMBC.

TECHNICAL SKILLS

AI AUTOMATION

Agent Design, Agentic Workflows, Multi-Agent Orchestration, Delegation Frameworks, Prompt, Context & Intent Engineering, Knowledge Base Design, Model Context Protocol (MCP), Claude Agent SDK, Crew.ai, Hermes Agent, OpenClaw, Low-Code Automation, n8n, Gemini,

AI INFRASTRUCTURE & PLATFORM

Self-Hosted LLM Inference (Ollama), Model Routing & Gateways (OmniRoute, OpenRouter, Nous Research), Vector Databases (Qdrant), RAG Pipelines, MCP Server Operations, Agent Memory Architecture, Secrets Management & Machine Identity (Infisical), AI

ERP & BUSINESS SYSTEMS

SyteLine, SAP S/4HANA, QAD, Sage, QuickBooks, Workwise, MRM, AMS

Claude, ChatGPT, LangChain, AutoResearch AI

Governance, ROI & Returns Modeling for Agent Workloads

DATA & ANALYTICS

Microsoft Fabric, Power BI, Tableau, SQL, Python, R, DAX Studio, KNIME, Power Query

CRM & MARTECH

Salesforce, HubSpot, Hybris, GA4

PROGRAM MANAGEMENT

Jira (Global Admin), Confluence, Agile, Scrum, Lean Six Sigma, Post-Merger Integration, Change Management

WORK MANAGEMENT

Notion, Visio, MS Office, Adobe Suite, draw.io, SOLIDWORKS

WORKFLOWS, SKILLS & AGENTS I OPERATE

AGENTS — 17 DEFINED

Named agents with declared model, tools and scope, each written for one job: researcher, project manager, backend and frontend engineer, test verifier, validator, brand reviewer, and four adversarial reviewers that exist to refute rather than confirm. I write the brief and check the result; the agents do the building.

SKILLS — 44 ROUTED

Reusable procedures a person or an agent can invoke by name, each with its trigger, its steps and the command that proves each step. A capability with no trigger is treated as absent, so adding one means adding the thing that fires it in the same change.

WORKFLOWS — 14 CI, 22 REFUSING HOOKS

Fourteen continuous-integration workflows plus twenty-two session hooks that refuse work rather than warn about it: no deliverable closes without a ledger entry, no claim stands without the file it cites, no commit lands with a credential shape in it.

SELF-PROVING INSTRUMENTS — 273

Every instrument answers --selftest and its selftests are refusals, not reports. 207 of them invoke their own entry point rather than only their internals, measured by a census script rather than counted by hand. Mutation

testing proves a control can actually fail before it is trusted.

PROFESSIONAL EXPERIENCE

Syrup Labs AI (syrp.ai)

Feb 2026 to Present

Founder, AI Automation Consultant

Chadds Ford, PA

Founder and AI Automation Consultant. AI automation consultancy building and deploying bootstrapped agentic systems, lead capture workflows, and customer support automation for small and local businesses. Built AI automation systems for retail, hospitality, and local services use cases.

- Led end-to-end program delivery for AI automation builds, from scoping through deployment: scoped, priced, built, tested, and rolled out agentic systems with ROI tied to admin time saved and lead capture volume.
- Built and internally validated AI concierge, intake, and lead-routing systems on Syrup Labs' own infrastructure, ahead of a scoped $15K fixed-price pilot with a prospective client.
- Built parent/subagent architectures grounded in business-specific knowledge bases, with prompt guardrails, FAQ handling, escalation logic, and intent separation between support and qualified lead triggers.
- Implemented low-code automation pipelines using HubSpot, n8n, Google Sheets, and Gmail; eliminating 10+ hours of manual admin per week per deployment.
- Designed and shipped Mission Control, an agent orchestration platform with its own integrations layer and settings panels: fleet and task boards, agent registry, cost and usage telemetry, knowledge search, and operator controls built in TypeScript and Next.js. Now 161 API routes, 276 unit test files, 72 Playwright end-to-end specs and 14 CI workflows, with 566 pull requests merged between August and October 2026.
- Architected and operate the firm's own agent platform: multi-agent orchestration across four machines with self-hosted model inference and routing, a vector-backed memory layer, MCP tool servers, machine-identity secrets rotation, and a CI estate with required checks and a self-hosted runner.
- Built governance into the platform rather than around it: refusing checkers that block unsafe actions, append-only audit receipts for every agent call, consent gates on personal data, and delegation rules that separate the agent doing work from the agent judging it.
- Translated ambiguous business goals into structured automation roadmaps; managed full engagement lifecycle from discovery through phased rollout and stakeholder onboarding.
- Operate a self-hosted stack of five containerized services continuously since June 2026 at $26 per month. Verified disaster recovery across 841 GB: encrypted backup with a SHA-256 check, two verified

snapshots, and a completed cold restore.

- Audited my own research-grading pipeline, found the judge was returning a passing template regardless of input, and closed the pipeline rather than ship it.

DwyerOmega

AI and Enterprise Transformation Manager

Jan 2024 to Feb 2026

Chadds Ford, PA

AI and Enterprise Transformation Manager. PE-backed industrial sensors, instrumentation, and semiconductor manufacturer; enterprise transformation spanning ERP/MRP, AMS, CRM/MarTech, and enterprise BI across Operations, Finance, Commercial, IT, Supply Chain, and PE stakeholders.

- Led AI and enterprise transformation programs across 6 acquisitions ($800M+), integrating ERP/MRP (SAP S/4HANA, SyteLine, QAD), AMS, CRM/MarTech, and enterprise BI while reporting directly to PE stakeholders and C-suite leadership.
- Applied AI and large language models across operational workflows including documentation generation, requirements analysis, data reconciliation automation, and vendor performance reporting, delivered at scale across post-merger integration environments.
- Managed program and project execution across cross-functional teams spanning Operations, Finance, Commercial, IT, and Supply Chain; standardized KPIs and integration playbooks for repeatable M&A execution.
- Owned enterprise reporting governance and reconciliation across global orders, sales, pricing, quality, and inventory; supported executive and PE decisions tied to pricing, margin, and working capital.
- Oversaw 15-20 resource offshore vendor team: day-to-day tasking, prioritization, progress reporting, and delivery performance tracking.
- Served as Global Admin for Jira and Confluence; deployed AI-assisted knowledge management in Confluence to improve documentation quality and governance scalability.
- Led greenfield implementation planning for enterprise AMS and Microsoft Fabric; coordinated cross-functional discovery, harmonization, and planning with external consulting partner.
- Built repeatable data-forensics workflow using Power Query, DAX Studio, and KNIME to reconcile large datasets and validate executive reporting when data integrity was disputed.

Senior Automation Engineer

Apr 2021 to Jan 2024

Swedesboro, NJ

Automation, analytics, and continuous improvement in PE-backed sensor and wire manufacturing; 3 consecutive Top Performer of the Year awards.

- Designed and executed $2M+ in automation and robotics-driven capital projects, improving throughput, reducing labor and overhead costs, and supporting sensor performance; served as cross-functional lead coordinating engineering, operations, and supply chain.
- Applied analytics and predictive methods to planning and inventory; reduced on-hand inventory by more than 60% across a $20M+ inventory base while maintaining internal service levels and improving delivery performance.
- Led Lean and continuous improvement initiatives including standardized work design, root-cause problem solving, and Kaizen events; drove sustained improvements in scrap, uptime, and changeover.
- Built KPI dashboards with operations, engineering, and supply chain leadership to improve executive visibility into output, scrap, and operational constraints.

Copperweld

Nov 2020 to Apr 2021

Production Manager

Pennsauken, NJ

Operations leadership across 40+ employees and 14+ lines in a 24/7 facility. Role ended due to facility closure following corporate reorganization.

- Managed 40+ employees across 14+ production lines in a 24/7 environment, driving safety, quality, and delivery performance.
- Contributed to $3M+ in annualized labor savings through workflow redesign, line balancing, and reduced overtime.

Selit North America, Inc.

Oct 2018 to Nov 2020

Process & Manufacturing Engineer

Plattsburgh, NY

24/7 foam extrusion and recycling operations for flooring OEM underlayment. Key contributor to OSHA SHARP certification.

- Spearheaded startup of a $20M sister facility near Atlanta, GA, including full plant design, equipment procurement, process development, and production line commissioning from greenfield.
- Served as day-to-day technical lead for a 30+ FTE production team across safety, quality, process, and customer issues.

Toray Plastics

May 2017 to May 2018

Process Engineer

Front Royal, VA

Foam extrusion for global luxury automotive OEMs; OSHA Star-certified site.

- Optimized final stages of the foam expansion process, increasing line speeds by an average of 60 meters per hour and generating $1M+ in annualized revenue impact.

EDUCATION

Villanova University

Master of Business Administration (MBA)

Strategic Management, Analytics & AI/ML

Villanova, PA · 2023-2025 · GPA 4.0

Beta Gamma Sigma Honor Society

University of Maryland, Baltimore County

Bachelor of Science, Chemical Engineering

Catonsville, MD · 2010-2016

CERTIFICATIONS

APICS CSCP

Atlassian Agile Project Management Professional

Google Cybersecurity Professional

Lean Six Sigma Green Belt, Drexel University

ISO 9001 Foundation

Value Analysis & Value Engineering, Lean Focus

VOLUNTEERING & ENTREPRENEURIAL

National Multiple Sclerosis Society

Rider & Volunteer, Bike MS · 2003-Present

22+ year participant in Bike MS; raised more than $30,000 for MS research to date; consistently among top fundraisers.

→ View my fundraising page

Little Thistle Farms

Co-Founder & Operator · 2023-Present

2-acre organic heirloom farm LLC in Chadds Ford, PA; garlic, flowers, and herbs, plus heirloom canned and pickled goods; 1.1K Instagram following; active local business partnerships.

→ littlethistlefarms.com